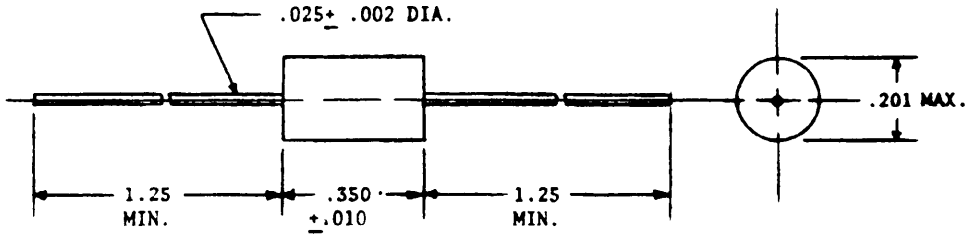
MILITARY SPECIFICATION SHEET

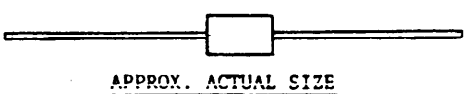
CAPACITORS, FIXED, MICA DIELECTRIC, ESTABLISHED RELIABILITY

STYLE CMR16

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-39001.





APPROX. ACTUAL SIZE



CIRCUIT DIAGRAM

INCHES	MM
.002	.05
.010	.25
.025	.64
.201	5.11
.350	8.89
1.25	31.8

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Style CMR16 capacitors.

FSC 5910

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REQUIREMENTS:

Dimensions: See figure 1.

Case material: Tubular, molded plastic, encapsulated.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC rated voltage: 100 V, 300 V, or 500 V, as shown in table I.

Operating temperature range: -55° to +125°C.

Characteristic: C,E or F, See table I.

Failure rate level: L,M,P,R and S, see table I.

Dielectric withstanding voltage (DWV): Method 301 of MIL-STD-202: 200 percent of rated dc voltage for not less than 1 second nor more than 5 seconds.

Barometric pressure: Method 105, condition D (100,000 ft), of MIL-STD-202. Test potential -- 150 percent of rated voltage, not to exceed 500 VDC.

Insulation resistance (IR): Method 302, test condition A, of MIL-STD-202: (Minimum requirements):
At 25°C: 100 Gigaohms.

At 125°C: 10 Gigaohms.

Capacitance (Cap.): Method 305 of MIL-STD-202.

Dissipation factor (DF): See MIL-C-39001.

High voltage stabilization: 200 percent of rated dc voltage for 48 hours at a temperature of 125°C +5 -0°C.

Solderability: Method 208 of MIL-STD-202.

Vibration, high frequency: Method 204 of MIL-STD-202.

Temperature coefficient and capacitance drift: Within the limits specified for the applicable characteristic.

Thermal shock and immersion:
Thermal shock: Method 107, condition A, MIL-STD-202, except high temperature shall be +125°C.

Immersion: Method 104, condition B, of MIL-STD-202.
DWV - Meet initial requirement.
IR - At least 30 percent of initial requirement.
ΔCap. - Shall not exceed ±1.5 percent of intital measurement or 1 pF, whichever is greater.
DF - Shall meet initial requirement.

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Shock, specified pulse: Method 213, condition I (100 g(pk), sawtooth), of MIL-STD-202.

Terminal strength:

Pull test - In accordance with MIL-STD-202, method 211, test condition A, 5 pounds pull applied in the direction of the leads for at least 5 seconds.

Twist test - In accordance with MIL-STD-202, method 211, test condition E, 5 alternate 360° rotations.

Moisture resistance: Method 106 of MIL-STD-202, except 20 cycles.

DWV - Meet initial requirement.
IR - At least 25 percent of initial requirement.
ΔCap. - Shall not exceed ±1.5 percent of initial measurement or 1 pF, whichever is greater.
DF - Not greater than 150 percent of initial requirement.

Life:

Forty-eight hours, minimum, at $-55^{\circ}\ {}^{+0^{\circ}}_{-2^{\circ}}C$ immediately followed by qualification or quality conformance test, as applicable.

Qualification test: 2,000 hours at +125°C with 100 percent or 150 percent of rated dc voltage applied, as applicable.

DWV - Meet initial requirement.
IR - Meet initial requirement.
ΔCap. - Shall not exceed ±1.5 percent of initial measurement of 1 pF, whichever is greater.
DF - Not greater than 150 percent of initial requirement.

Quality conformance test (240- and 2000-hour): 240 hours (L FR level) and 2000 hours (M, P, R, and S FR levels) at +125°C and 150 percent of rated dc voltage; requirements as specified for qualification.

Extended life (L FR level only):

Qualification test: $48{,}000\ {}^{+96}_{-0}$ hours at +125°C and rated dc voltage; requirements as specified for qualification.

Quality conformance test: 1,760 hours at applicable high test temperature and 150 percent of rated dc voltage; requirements as specified for qualification.

Fungus: Method 508 of MIL-STD-810 or manufacturer's equivalent.

Resistance to solvents: Method 215 of MIL-STD-202.

Resistance to soldering heat: Method 210, Test condition E, of MIL-STD-202.

IR - Meet initial requirement.
ΔCap. - Shall not exceed ±1.5 percent of of initial measurement or 1 pF, whichever is greater.
DF - Meet initial requirement.

Marking: Capacitors shall be marked as indicated in figure 2.

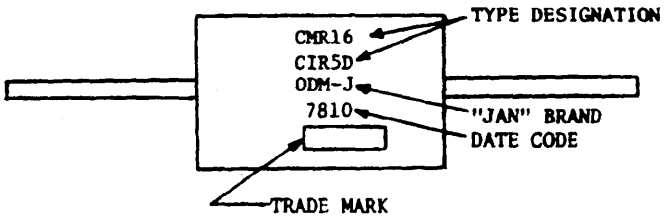


FIGURE 2. Marking example.

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TABLE I. Type designations

STYLE CMR16			
Type designation 1/	Capacitance value	DC rated voltage	Capacitance tolerance available
	(pF)	(volts)	
CMR16C1R0D0D-	1.0	500	D
CMR16C1R5D0D-	1.5		
CMR16C2R0D0D-	2.0		
CMR16C2R5D0D-	2.5		
CMR16C3R0D0D-	3.0		
CMR16C3R5D0D-	3.5		
CMR16C4R0D0D-	4.0		
CMR16C4R5D0D-	4.5		
CMR16C5R0D0D-	5.0		
CMR16C6R0D0D-	6.0		
CMR16C7R0D0D-	7.0		
CMR16C8R0D0D-	8.0		
CMR16C9R0D0D-	9.0		
CMR16C100D0D-	10		D
CMR16C110-0D-	11		D,J
CMR16C120-0D-	12		
CMR16C150-0D-	15		
CMR16C180-0D-	18		
CMR16E200-0D-	20		
CMR16E220-0D-	22		
CMR16E240-0D-	24		D,J
CMR16E270-0D-	27		G,J
CMR16E300-0D-	30		
CMR16E330-0D-	33		
CMR16E360-0D-	36		
CMR16E390-0D-	39		
CMR16E430-0D-	43		
CMR16E470-0D-	47		G,J
CMR16E510-0D-	51		F,G,J
CMR16E560-0D-	56		
CMR16E620-0D-	62		
CMR16E680-0D-	68		
CMR16E750-0D-	75		
CMR16E820-0D-	82		
CMR16F910-0D-	91		
CMR16F101-0D-	100		
CMR16F111-0D-	110		
CMR16F121-0D-	120		
CMR16F131-0D-	130		
CMR16F151-0D-	150		
CMR16F161-0D-	160		
CMR16F181-0D-	180		
CMR16F201-0D-	200		
CMR16F221-0D-	220		
CMR16F241-0D-	240	500	
CMR16F271-0C-	270	300	
CMR16F301-0C-	300		
CMR16F331-0C-	330		
CMR16F361-0C-	360		
CMR16F391-0C-	390	300	
CMR16F431-0A-	430	100	
CMR16F471-0A-	470		
CMR16F511-0A-	510		
CMR16F561-0A-	560		
CMR16F621-0A-	620		
CMR16F681-0A-	680	100	F,G,J

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1/ Complete type designation will include additional letter symbols to indicate the following, where applicable:
Capacitance tolerance: D (±.5 pF); F (±1%); G (±2%); or J (±5%).

Failure rate level: L (2.0%); M (1.0%); P (0.1%); R (0.01%); or S (0.001%).

Packaging:

Capacitors will be furnished in tape and reel packaging when so specified in the ordering data.

Application Notes:

1. Where applicable, these capacitors can be used as replacements for CM[1] capacitors listed in MIL-C-5/1.

2. Substitutability for failure rates and capacitance tolerances shall be as follows:

Failure Rate	Will Replace Failure Rate	Cap Tolerance	Will Replace Tolerance
S	R, P, M, L	D	J
R	P, M, L	F	G,J
P	M, L	G	J
M	L		

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Air Force - 85
DLA - ES

User activities:
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5910-1245)

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE MIL-C-39001/6 CAPACITORS, FIXED, MICA DIELECTRIC, ESTABLISHED RELIABILITY STYLE CMR16

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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